

Sri Lanka Institute of Information Technology



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Specializing in Data Science

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Year 3

Assignment 1

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1. Data Set Selection

1.1 Introduction

In modern banking systems, large amounts of transactional information are created day by day. This data contains customer information, account information, transaction details, loans, and card usage. However, such data is typically stored in OLTP systems optimized for real-time operations rather than analytical processing.

The project dwells upon the creation of a data warehouse application and implementation of a Banking Transaction dataset data warehouse. The system consolidates the raw data and converts data through ETL process and stores data in dimensional model to aid analytical queries and business intelligence.

1.2 Dataset Overview

The dataset used in this project is a comprehensive banking dataset stored in a CSV file. It holds various customer activities and financial transactions attributes. This data set is an OLTP system since it includes transactional data, which include transaction, loan, and account operations.

The following logical entities are contained in the dataset:

- Customer – personal and contact details
- Account – account type and balance
- Transaction – transaction details
- Loan – loan information
- Card – credit/debit card data
- Feedback – customer feedback

The data is complex and voluminous enough to be used to model banking activities in the real world.

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Source Link: [Banking-Dataset/Comprehensive_Banking_Database.csv](#) at main · ahsan084/Banking-Dataset · GitHub

The data initially came in the form of one CSV file. To model an OLTP system in real-world, the dataset was logically split into various entities and loaded into SQL server.

- SrcCustomer
- SrcAccount
- SrcTransaction
- SrcLoan
- SrcCard

ETL. This resembles a multi-source setting that integrates flat file sources and relational database sources

1.4 OLTP Characteristics

The dataset represents an OLTP system because:

- Data is highly normalized across multiple tables
- It supports real-time transactional operations
- Frequent insert, update, and delete operations occur
- Each record represents a business transaction
- Relationships are maintained using primary and foreign keys

Therefore, the existing data is appropriate to be converted into a data warehouse.

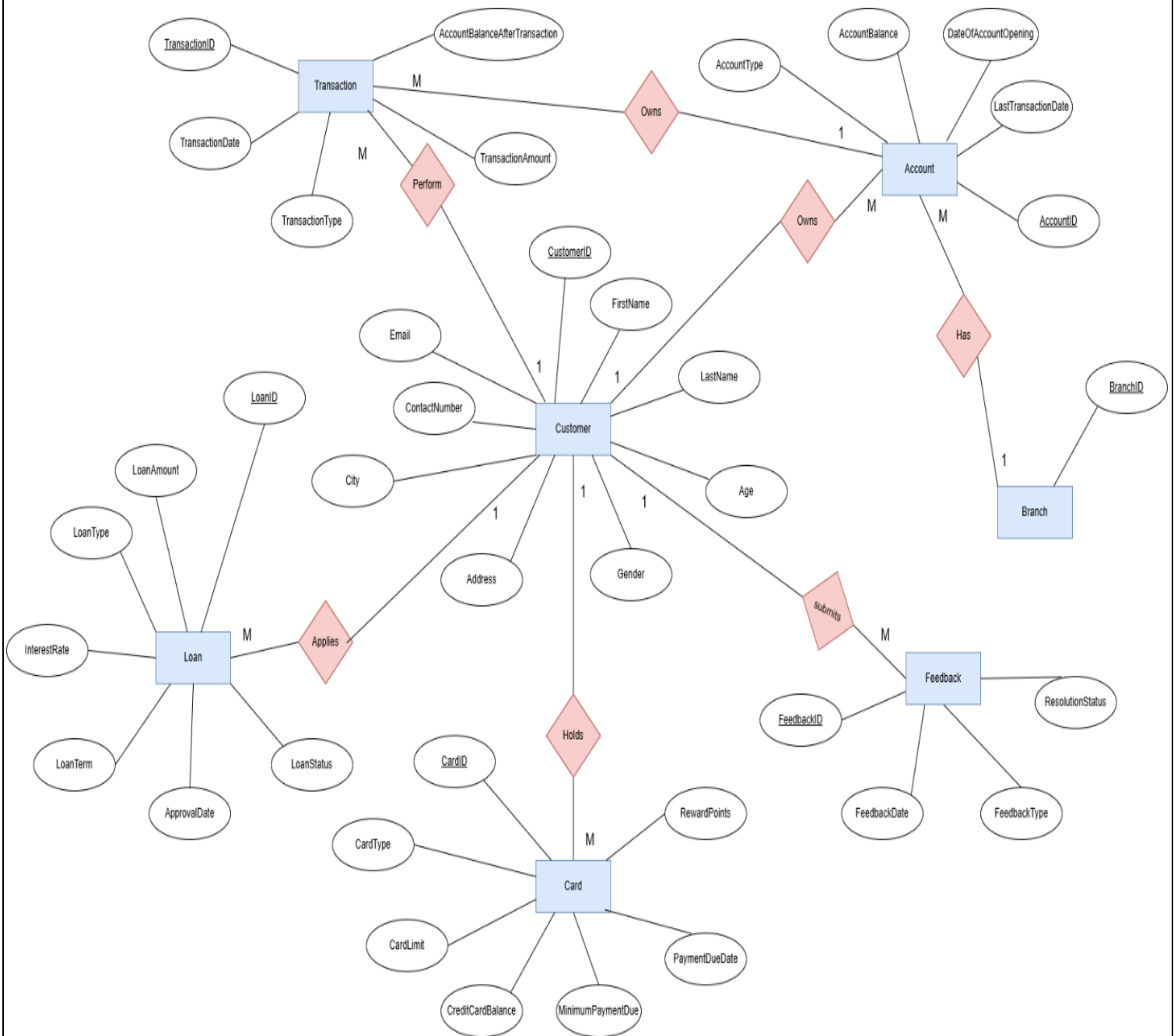
1.5 Suitability for Data Warehousing

The dataset is suitable for data warehousing because:

- Real-world banking scenario
- Large volume of data
- Multiple related entities
- Ability to perform ETL transformations
- Support dimensional modeling
- Ability to create analytical conclusions

This can be best used in business intelligence applications.

1.6 ER Diagram



2. SOURCE DATA DESCRIPTION

2.1 CSV Data Source

The primary data source is a CSV file containing all banking data. The file includes attributes related to customers, transactions, loans, accounts, and cards.

This file simulates raw data collected from external systems.

2.2 Source Database Design

The CSV file was transformed into a structured database name “Banking_SourceDB.”

Table Name	Description
SrcCustomer	Customer information
SrcAccount	Account details
SrcTransaction	Transaction data
SrcLoan	Loan data
SrcCard	Card details

 dbo.SrcAccount
 dbo.SrcCard
 dbo.SrcCustomer
 dbo.SrcLoan
 dbo.SrcTransaction

Figure 2.2.1: Tables in Banking_SourceDB

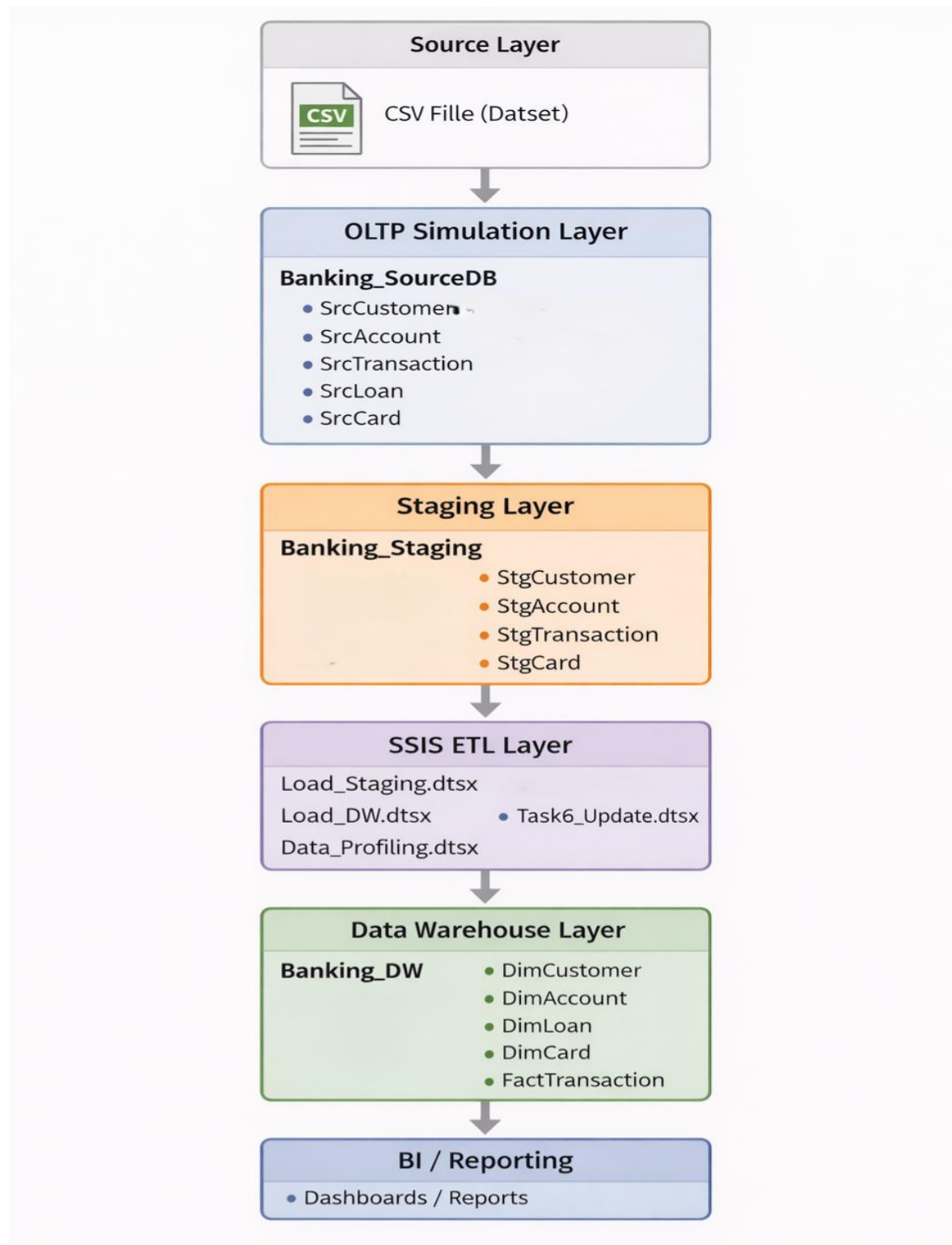
This database acts as the OLTP system for the ETL process.

	Customer_ID	First_Name	Last_Name	Age	Gender	Contact_Number	Email
1	1	Joshua	Hall	45	Male	19458794854	joshua.hall@kag.com
2	2	Mark	Taylor	47	Female	19458794855	mark.taylor@kag.com
3	3	Joseph	Flores	25	Female	19458794856	joseph.flores@kag.com
4	4	Kevin	Lee	52	Other	19458794857	kevin.lee@kag.com
5	5	Linda	Johnson	68	Other	19458794858	linda.johnson@kag.com
6	6	Charles	Torres	52	Male	19458794859	charles.torres@kag.com
7	7	Joshua	Garcia	43	Male	19458794860	joshua.garcia@kag.com
8	8	Christopher	Baker	47	Male	19458794861	christopher.baker@kag.com
9	9	Daniel	Moore	52	Male	19458794862	daniel.moore@kag.com
10	10	William	Robinson	55	Other	19458794863	william.robinson@kag.com

Figure 2.2.1: Query result in SrcCustomer(Banking_SourceDB)

3. SOLUTION ARCHITECTURE

3.1 Architecture Diagram



3.2 Component Description

The architecture consists of three layers:

1. Source Layer – CSV dataset and SQL Server source tables
2. Staging Layer – Temporary storage for cleaning & transformation
3. Data Warehouse Layer – Dimensional model for analysis

ETL processes are implemented using SSIS to move data between layers.

Source Layer:

- CSV File (raw dataset)

OLTP Simulation Layer:

- Banking_SourceDB

Staging Layer:

- Banking_Staging (temporary cleaned data)

ETL Layer:

- SSIS packages for transformation

Data Warehouse Layer:

- Banking_DW (star schema)

BI Layer:

- Reporting and analysis tools

4. DATA WAREHOUSE DESIGN

4.1 Dimensional Model

Fact Table:

- FactTransaction

The FactTransaction table stores measurable transactional data such as transaction amount and processing time. It connects to dimension tables such as Customer, Account, and Date through foreign keys.

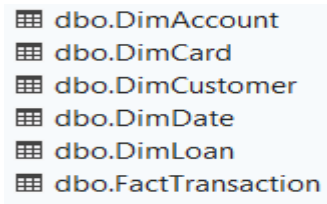
Derived attributes such as txn_process_time_hours were calculated during ETL.

Column Name	Data Type	Allow Nulls
TransactionSK	int	☐
CustomerSK	int	☒
AccountSK	int	☒
LoanSK	int	☒
CardSK	int	☒
DateKey	int	☒
TransactionID	int	☒
TransactionType	nvarchar(50)	☒
TransactionAmount	float	☒
AccountBalanceAfterTransaction	float	☒
InsertDate	datetime	☒
accm_txn_create_time	datetime	☒
accm_txn_complete_time	datetime	☒
txn_process_time_hours	float	☒
		☐

Figure 4.1.1: Fact Table Structure

Dimension Tables:

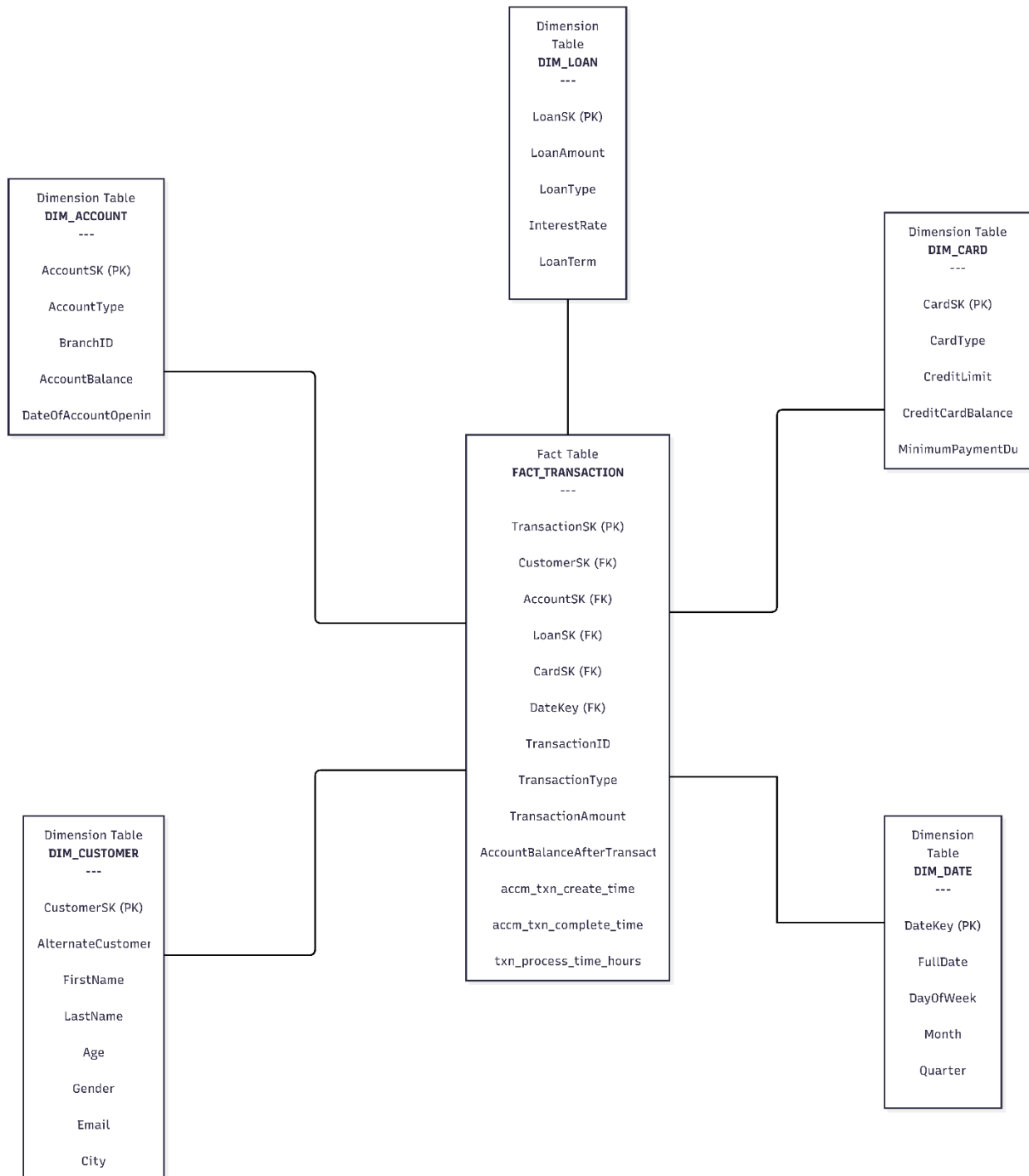
- DimCustomer
- DimAccount
- DimLoan
- DimCard
- DimDate



- dbo.DimAccount
- dbo.DimCard
- dbo.DimCustomer
- dbo.DimDate
- dbo.DimLoan
- dbo.FactTransaction

Figure 4.1.1: Banking_DW Tables' Structure

4.2 Star Schema Diagram



4.3 Design Assumptions

- Star schema used for simplicity and performance
- Fact table is a transaction-level data storage
- Dimensions give descriptive attributes
- Surrogate keys used for relationships
- Derived columns calculated during ETL

Grain:

The grain of the FactTransaction table is one record per transaction.

Measures:

- TransactionAmount
- txn_process_time_hours

Slowly Changing Dimension: DimCustomer is used SCD Type 2 to trace the changes in the past.

5. ETL DEVELOPMENT

5.1 SSIS Packages

The ETL process was implemented using SSIS with the following packages:

Package Name	Purpose
Banking_Load_Staging.dtsx	Load data to staging
Banking_Load_DW.dtsx	Load data to DW
Task6_Fact_Update.dtsx	Update fact table
DataProfiling.dtsx	Data quality analysis

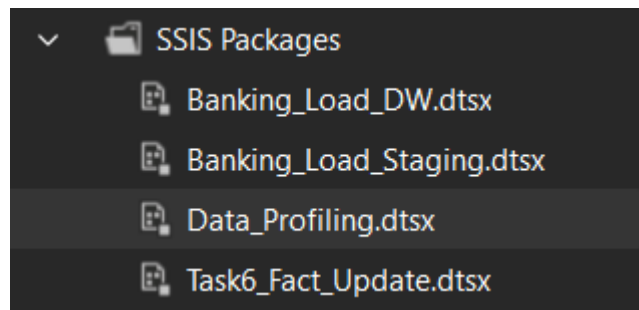


Figure 5.1.1: SSIS package modules as seen in Solution Explorer.

5.2 Staging Load

Data was extracted from Banking_SourceDB and loaded into Banking_Staging using SSIS Data Flow Tasks.

1. Extract Customer Data
2. Extract Account Data
3. Extract Card Data
4. Extract Loan Data
5. Extract Transaction Data

6. Extract Customer Feedback (from a flat file)

Transformations included:

- Data type conversion
- Null handling
- Data cleaning



Figure 5.2.1: Control Flow in ' Banking_Load_Staging.dtsx' showing staging tasks.

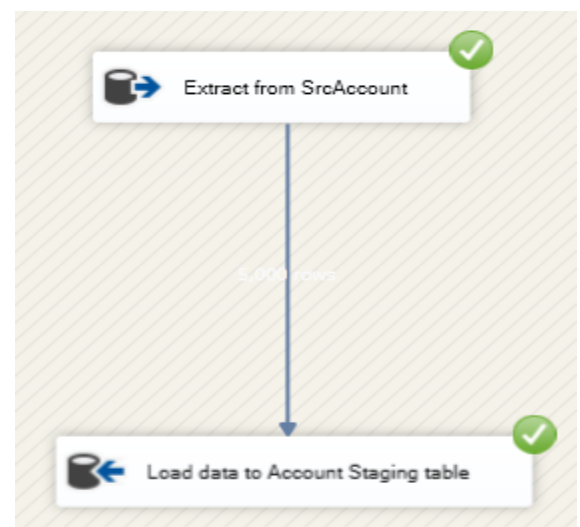
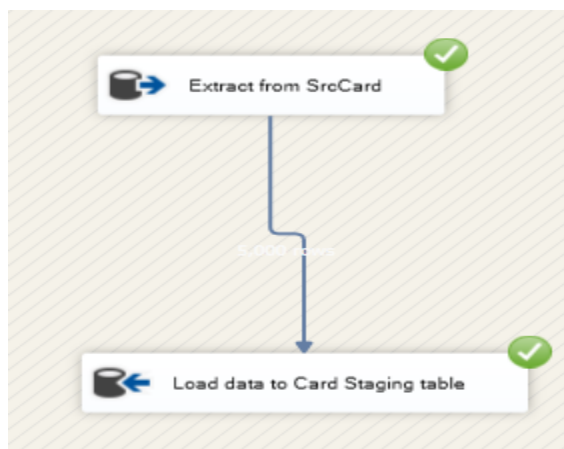
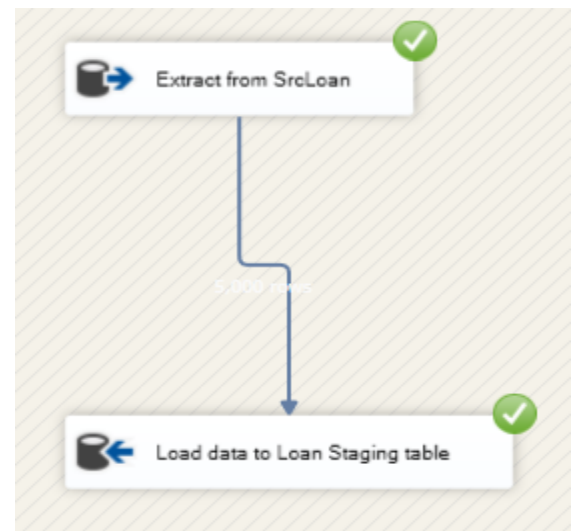
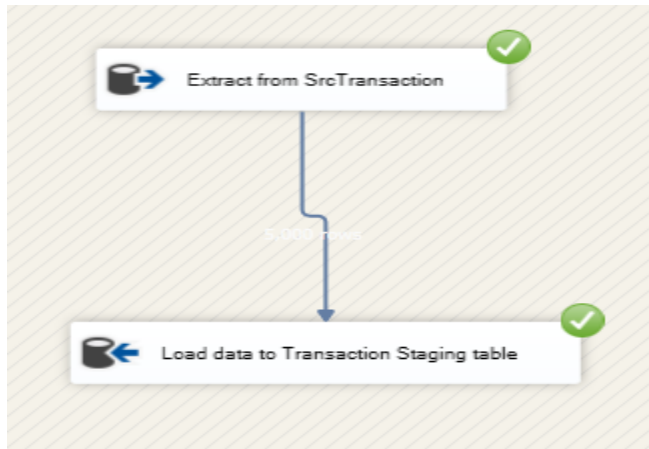
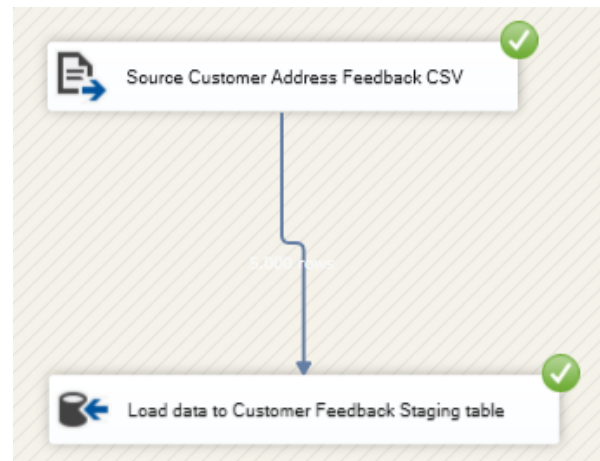
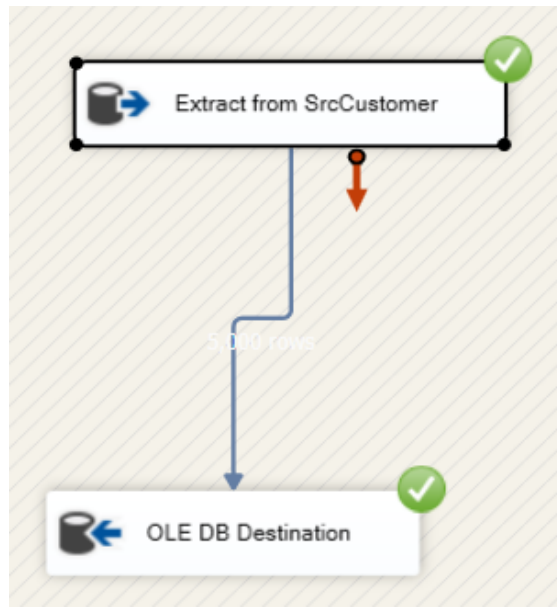


Figure 5.2.2: Example Data Flows from source to staging tables

5.3 Data Warehouse Load

Data from staging tables was loaded into dimension and fact tables.

Transformations included:

- Lookup transformations
- Surrogate key generation
- Data filtering

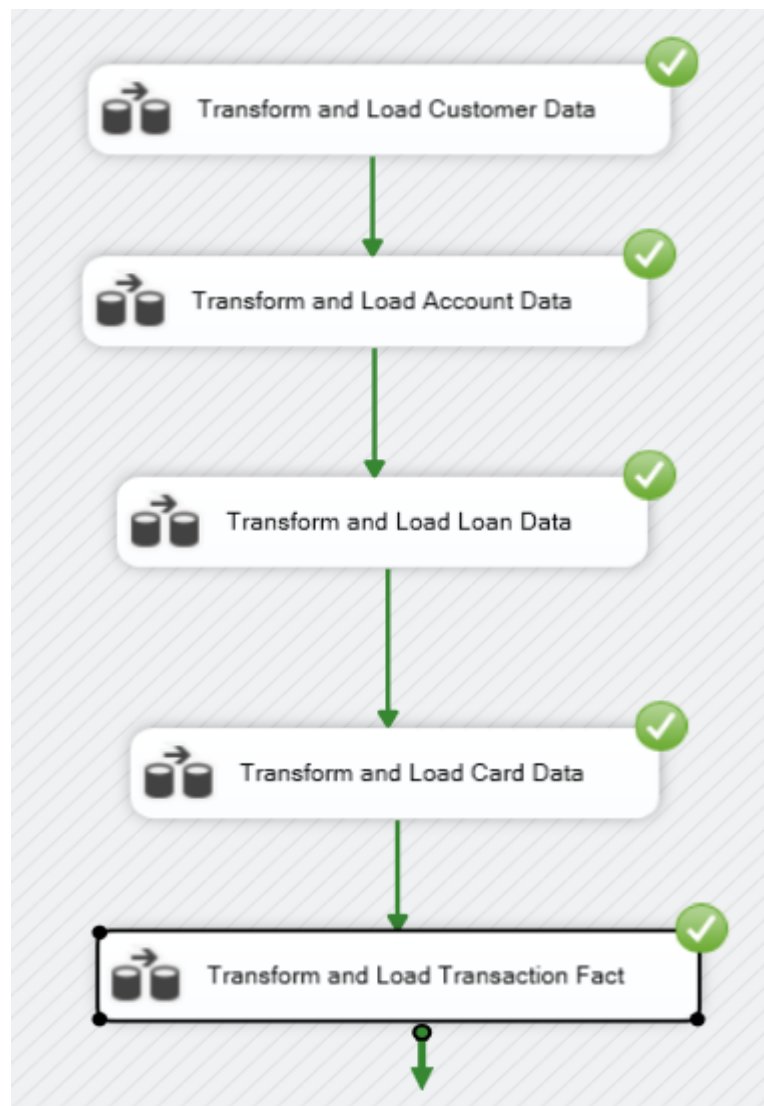


Figure 5.3.1: Control Flow in 'Banking_Load_DW.dtsx' showing dimensional and fact loading sequence.

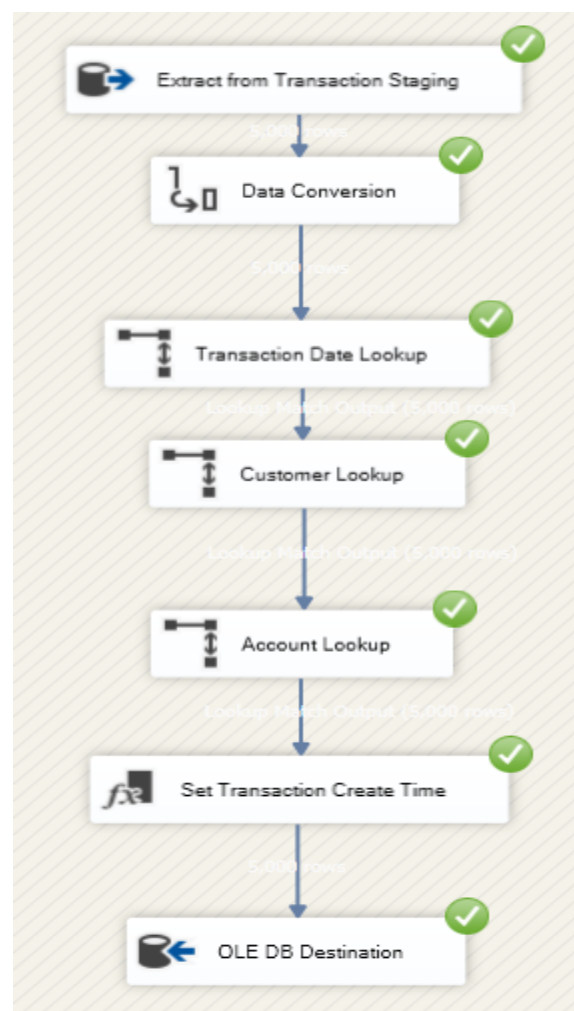
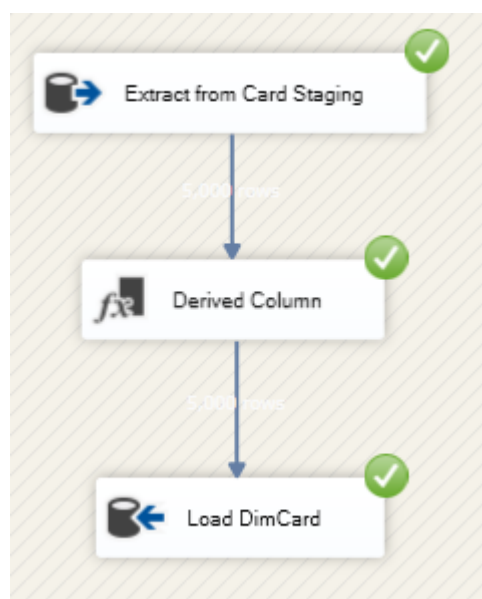
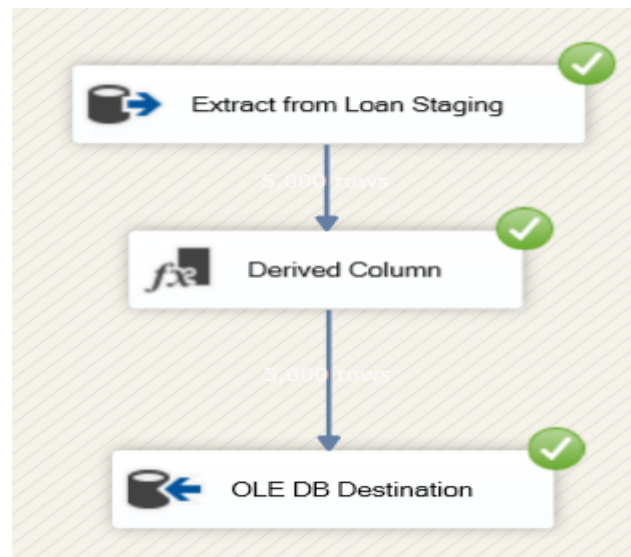
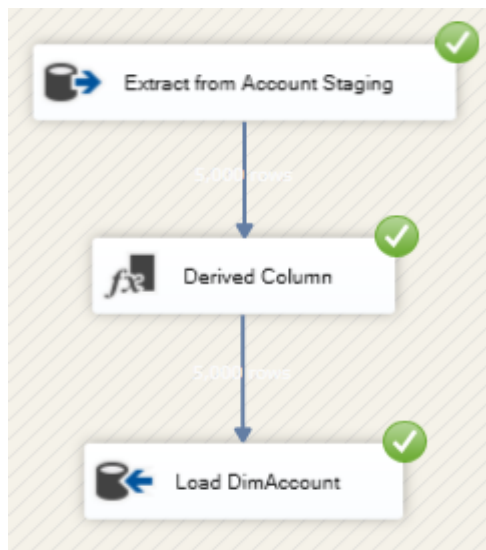


Figure 5.3.2: Loading the Transaction, Card, Account, Loan Dimension

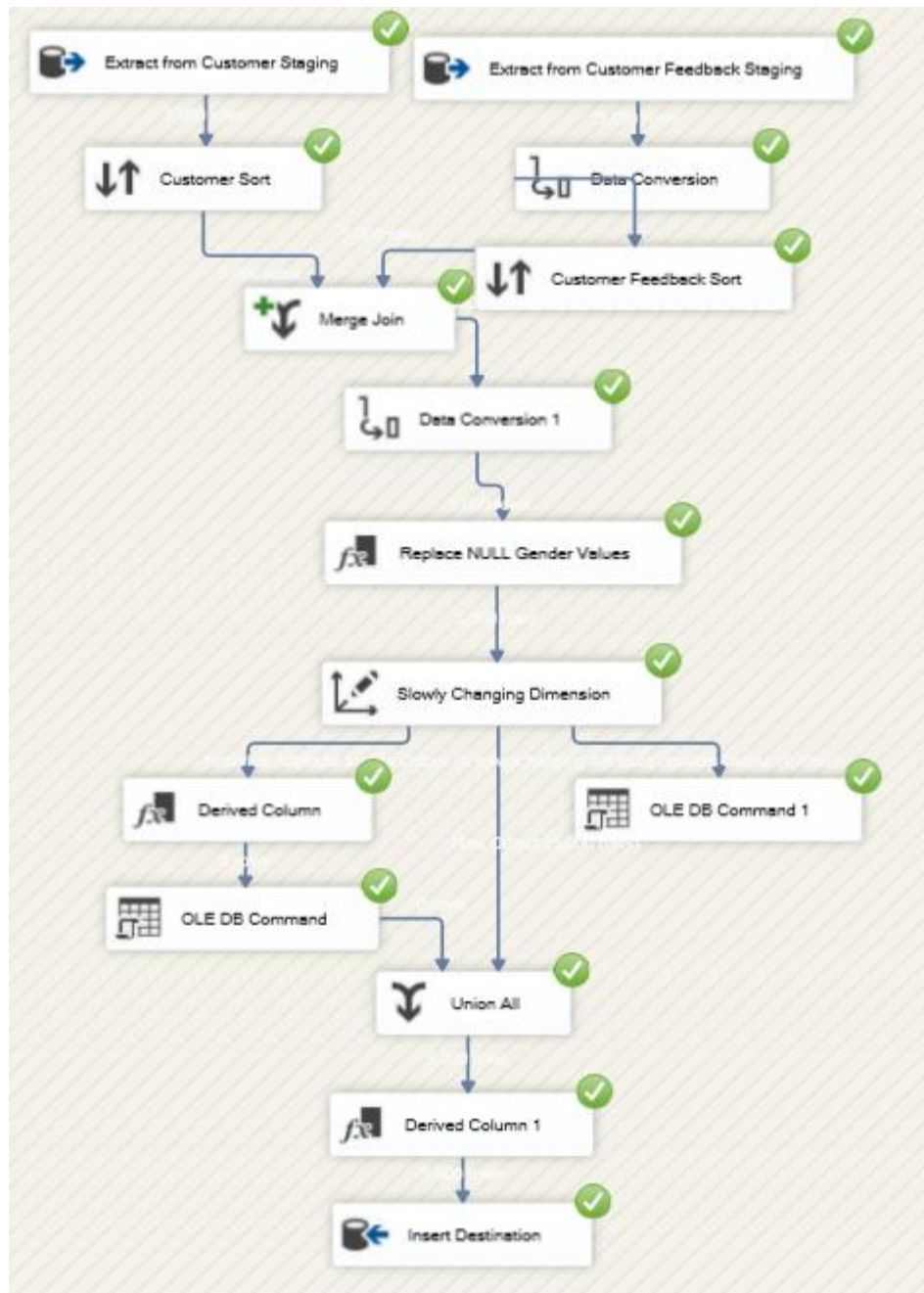


Figure 5.3.3: Loading the Customer Dimension

5.4 Fact Table Update

Task 6 involved updating the FactTransaction table to calculate transaction completion time and processing duration.

Columns used:

- accm_txn_create_time
- accm_txn_complete_time
- txn_process_time_hours

$\text{txn_process_time_hours} = \text{accm_txn_complete_time} - \text{accm_txn_create_time}$

Processing time was calculated using:

$\text{txn_process_time_hours} = \text{DATEDIFF}(\text{HOUR}, \text{accm_txn_create_time}, \text{accm_txn_complete_time})$

All null values were handled, and results were validated to ensure accuracy.

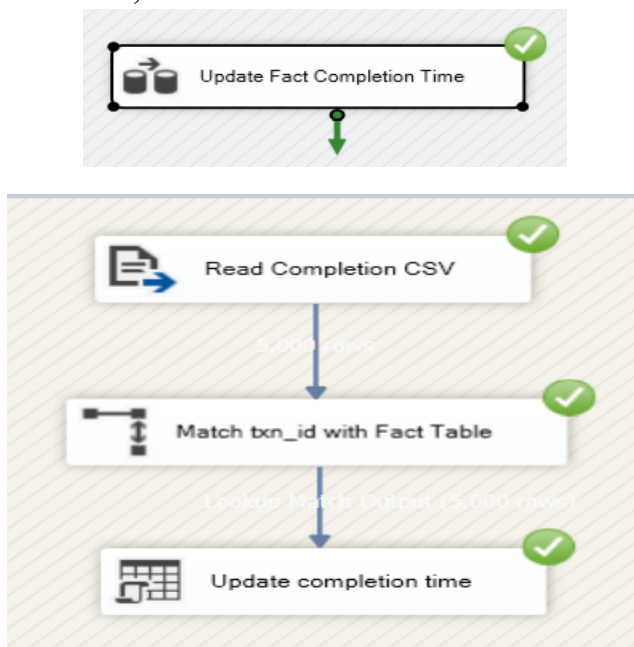


Figure 5.4.1: Control Flow in 'Task6_Fact_Update.dtsx'

5.5 Data Profiling

The SSIS Data Profiling Task was used to perform data profiling with the purpose of analyzing the quality of data in the staging database (Banking_Staging).

The process of profiling was used to stage the tables like StgCustomer to gain the following data characteristics like null values, column length and value distributions.

The following checks were conducted:

- Column Null Ratio Profile – to detect missing values
- Column Length Profile – to analyze data size consistency
- Column Value Distribution Profile – to understand data patterns

The storing of the profiling results was in an XML file and the results were viewed with the Data Profile Viewer. This step helped identify data quality issues before loading data into the data warehouse.

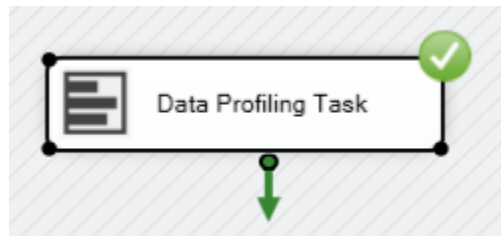


Figure 5.5.1: Data Flow in Data_Profiling.dtsx

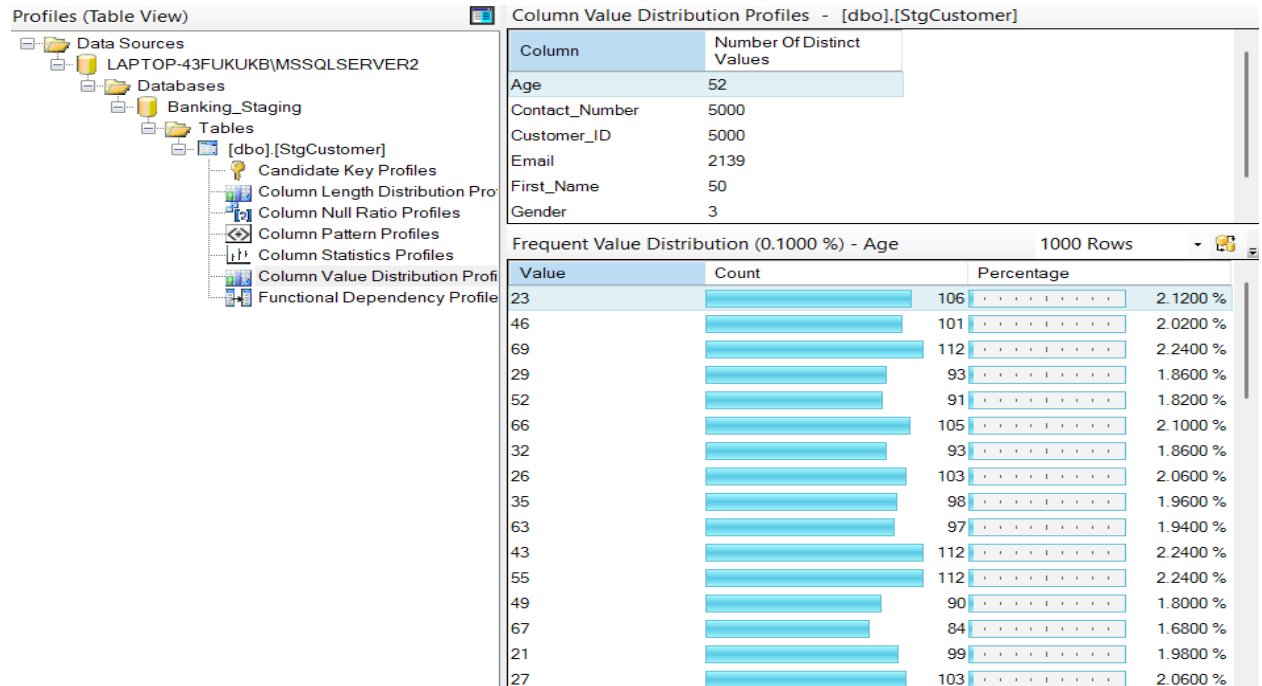
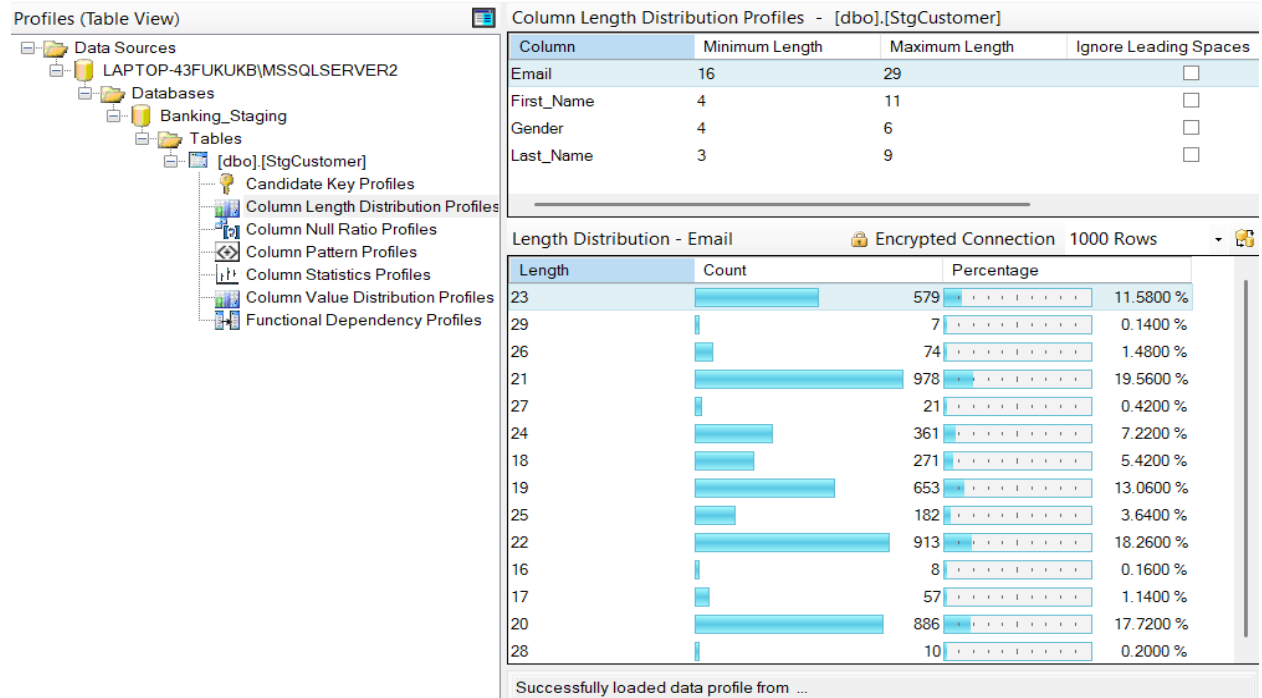


Figure 5.5.1: Profile Viewer Results for Data Profiling

5.6 Validation

Data validation was performed using SQL queries:

- Row counts verification
- Sample data checks
- Fact table verification

Row checking was done on staging tables to confirm that all source records were properly loaded. Sample queries were used, checking if dimension tables were transformed correctly. The validation of the fact table was done by verifying the calculated fields like the time taken to process the transactions. Further, a check of NULL was also done to check completeness of data.

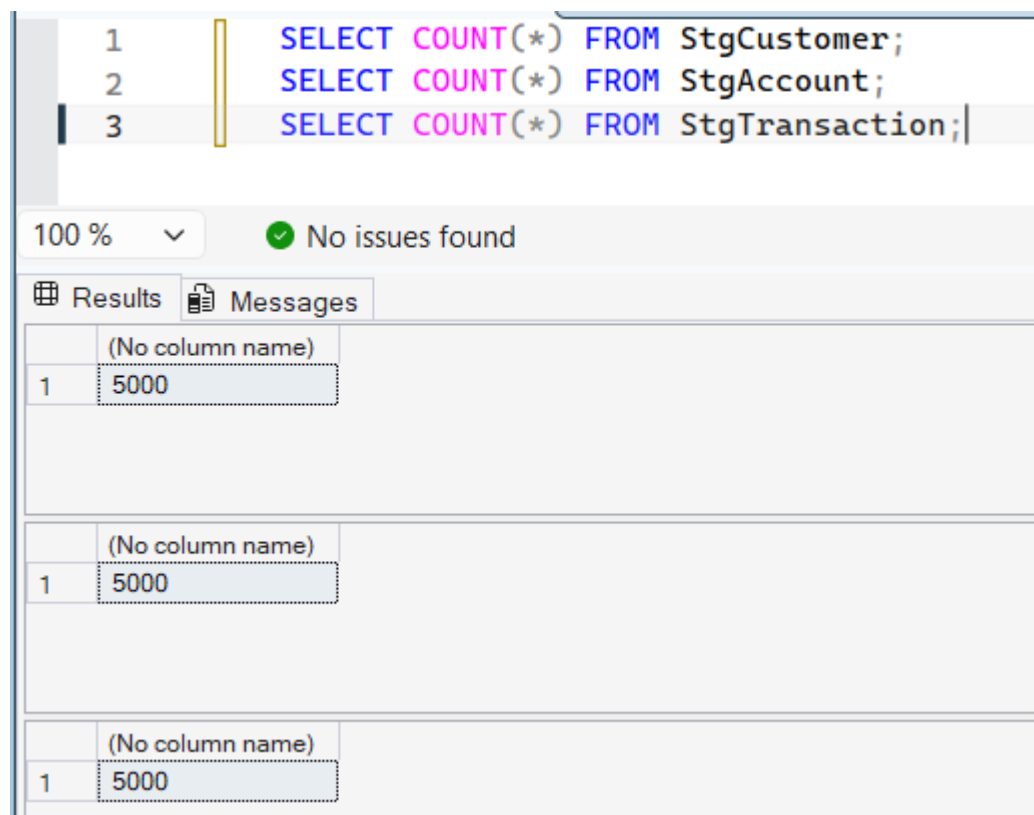


Figure 5.6.1: Staging Load Validation

1	SELECT TOP 10 * FROM DimCustomer;									
2	SELECT TOP 10 * FROM DimAccount;									

100 % ✓ No issues found Ln: 2, Ch: 33 (67 chars, 2 lines) TABS CRLF UTF-

	CustomerSK	AlternateCustomerID	FirstName	LastName	Age	Gender	ContactNumber	Email	Address	City	IsCurrent	InsertDate	ModifiedDate
1	1	1	Joshua	Hall	45	Male	19458794854	joshua.hall@kag.com	Address_1	Fort Worth	1	2026-03-27 12:30:53.417	2026-03-27 12:30:53.417
2	2	2	Mark	Taylor	47	Female	19458794855	mark.taylor@kag.com	Address_2	Louisville	1	2026-03-27 12:30:53.417	2026-03-27 12:30:53.417
3	3	3	Joseph	Flores	25	Female	19458794856	joseph.flores@kag.com	Address_3	Philadelphia	1	2026-03-27 12:30:53.417	2026-03-27 12:30:53.417
4	4	4	Kevin	Lee	52	Other	19458794857	kevin.lee@kag.com	Address_4	Oklahoma City	1	2026-03-27 12:30:53.417	2026-03-27 12:30:53.417
5	5	5	Linda	Johnson	68	Other	19458794858	linda.johnson@kag.com	Address_5	Phoenix	1	2026-03-27 12:30:53.417	2026-03-27 12:30:53.417
6	6	6	Charles	Torres	52	Male	19458794859	charles.torres@kag.com	Address_6	El Paso	1	2026-03-27 12:30:53.417	2026-03-27 12:30:53.417
7	7	7	Joshua	Garcia	43	Male	19458794860	joshua.garcia@kag.com	Address_7	Dallas	1	2026-03-27 12:30:53.417	2026-03-27 12:30:53.417
8	8	8	Christop...	Baker	47	Male	19458794861	christopher.baker@ka...	Address_8	Memphis	1	2026-03-27 12:30:53.417	2026-03-27 12:30:53.417

	AccountSK	AlternateCustomerID	AccountType	BranchID	AccountBalance	DateOfAccountOpening	LastTransactionDate	InsertDate	ModifiedDate
1	1	1	Current	43	1313.38000488281	2006-05-26	2023-04-23	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
2	2	2	Current	63	5988.4599609375	2006-03-02	2023-01-27	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
3	3	3	Current	82	8277.8798828125	2015-07-19	2023-01-01	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
4	4	4	Savings	41	7487.2099609375	2008-01-30	2023-08-02	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
5	5	5	Savings	9	6993.5498946875	2021-05-25	2023-03-27	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
6	6	6	Savings	20	9295.6396484375	2015-12-21	2023-06-04	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
7	7	7	Current	21	4005.71997070313	2022-07-07	2023-03-21	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
8	8	8	Savings	25	1458.44995117188	2022-12-14	2023-03-01	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
9	9	9	Current	66	1109.30004882813	2012-04-29	2023-11-05	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000
10	10	10	Current	30	9554.6201171875	2021-12-20	2023-02-05	2026-03-27 12:30:51.000	2026-03-27 12:30:51.000

Figure 5.6.2: Dimension Load Validation

1	SELECT TOP 20			
2	TransactionID,			
3	accm_txn_create_time,			
4	accm_txn_complete_time,			
5	txn_process_time_hours			
6	FROM FactTransaction;			

100 % ✓ No issues found

	TransactionID	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	1	2026-03-26 16:04:11.827	2026-03-28 14:31:21.000	46
2	2	2026-03-26 16:04:11.827	2026-03-29 15:31:21.000	71
3	3	2026-03-26 16:04:11.827	2026-03-30 16:31:21.000	96
4	4	2026-03-26 16:04:11.827	2026-03-31 17:31:21.000	121
5	5	2026-03-26 16:04:11.827	2026-03-27 18:31:21.000	26
6	6	2026-03-26 16:04:11.827	2026-03-28 19:31:21.000	51
7	7	2026-03-26 16:04:11.827	2026-03-29 20:31:21.000	76
8	8	2026-03-26 16:04:11.827	2026-03-30 21:31:21.000	101
9	9	2026-03-26 16:04:11.827	2026-03-31 22:31:21.000	126
10	10	2026-03-26 16:04:11.827	2026-03-27 23:31:21.000	31
11	11	2026-03-26 16:04:11.827	2026-03-29 00:31:21.000	56
12	12	2026-03-26 16:04:11.827	2026-03-30 01:31:21.000	81
13	13	2026-03-26 16:04:11.827	2026-03-31 02:31:21.000	106
14	14	2026-03-26 16:04:11.827	2026-04-01 03:31:21.000	131
15	15	2026-03-26 16:04:11.827	2026-03-28 04:31:21.000	36
16	16	2026-03-26 16:04:11.827	2026-03-29 05:31:21.000	61
17	17	2026-03-26 16:04:11.827	2026-03-30 06:31:21.000	86
18	18	2026-03-26 16:04:11.827	2026-03-31 07:31:21.000	111
19	19	2026-03-26 16:04:11.827	2026-04-01 08:31:21.000	136
20	20	2026-03-26 16:04:11.827	2026-03-28 09:31:21.000	41

Figure 5.6.3: Fact Table Validation

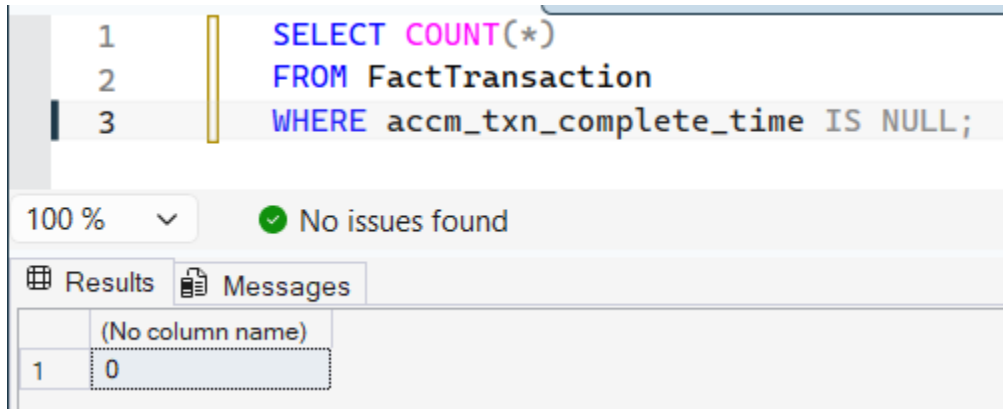


Figure 5.6.4: Null Check

6. CONCLUSION

This project was able to implement the entire data warehousing solution on a banking system.

The practical implementation of ETL processes and dimensional modeling is reflected in the transformation process of raw CSV data into an organized system of OLTP and the subsequent staging and data warehouse design.

The last system allows performing the efficient analytical processing and facilitates business intelligence, which makes data warehousing in contemporary data-driven settings important.